



# KNN

## CHAPTERS

- K-Nearest Neighbors (KNN)

**~1 contact hours**

## WHY THIS MODULE MATTERS

Describes KNN: a simple but often very effective ML ‘model’

# K-Nearest Neighbors



## Core Idea:

Classify a test example based on its similarity to known examples in the training set.

## Step-by-Step Process:

- **Measure Similarity**

Use a **distance function** (e.g., Euclidean, Manhattan) to compute how close the test example is to each training example.

- **Find Nearest Neighbors**

Identify the **K closest examples** in the training set.

- **Predict the Class**

- **Majority Voting:** Assign the most frequent class among the K neighbors.
- **Weighted Voting (optional):** Weight each neighbor's vote by its distance (closer = more influence).

# K-Nearest Neighbors

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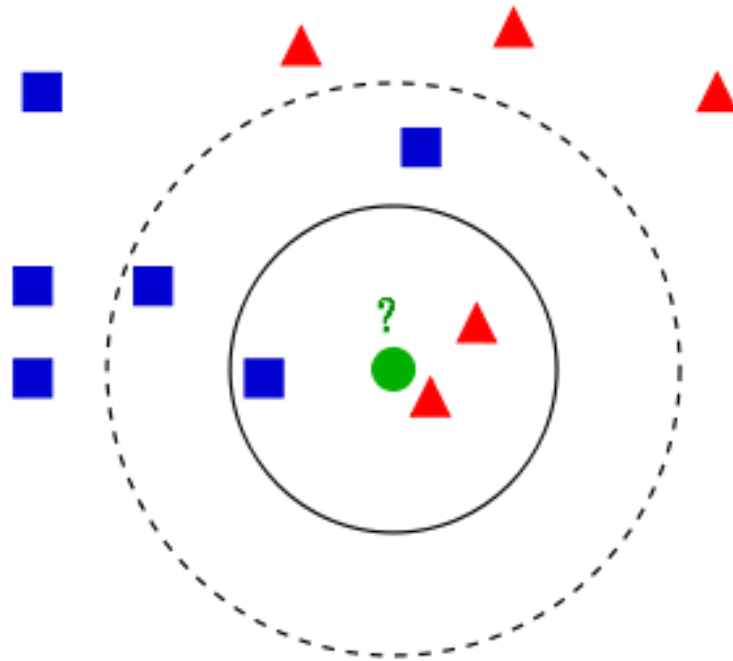
- Definition of metric (or distance)
- Given a set  $\mathcal{X}$ , a function  $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_0^+$  is a metric for  $\mathcal{X}$  if  $\forall x, y, z \in \mathcal{X}$  the following properties hold:
- Reflexivity  $d(x, y) = 0$  iff  $x = y$
- Symmetry  $d(x, y) = d(y, x)$
- Triangle inequality  $d(x, y) + d(y, z) \geq d(x, z)$

# K-Nearest Neighbors

Examples of metric

- Euclidean distance  $d(x, y) = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$
- Hamming distance  $d(x, y) = |\sum_{i=1}^d (x_i \neq y_i)|$

# K-Nearest Neighbors



K = 3  
K = 5  
???

[Image from Wikipedia]

# K-Nearest Neighbors

## Algorithm for K=1

For each test example  $S_j$

$\text{minD} = +\text{Infinity}$ ,  $\text{minLabel} = \text{NULL}$

    For each training example  $T_k$

$D = \text{distance}(S_j, T_k)$

        if  $D < \text{minD}$

$\text{minD} = D$

$\text{minLabel} = \text{label}[T_k]$

$\text{label}[S_j] = \text{minLabel}$

# K-Nearest Neighbors

Example with Euclidean distance

| Examples | X1   | X2  | X3  | Label |
|----------|------|-----|-----|-------|
| T1       | 0.5  | 1.2 | 2.3 | 1     |
| T2       | 0.2  | 1.1 | 3.5 | 0     |
| T3       | -0.6 | 0.7 | 5.4 | 1     |
| T4       | -0.1 | 0.5 | 1.2 | 0     |
| S1       | 0.6  | 1.1 | 2.1 | ?     |

$$d(S1, T1) = (0.5 - 0.6)^2 + (1.2 - 1.1)^2 + (2.3 - 2.1)^2 = 0.06$$

$$d(S1, T2) = 2.13$$

$$d(S1, T3) = 13.38$$

$$d(S1, T4) = 1.55$$

# K-Nearest Neighbors

Example with Euclidean distance

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| T3       | -0.6 | 0.7 | 5.4 | 1        |
| T4       | -0.1 | 0.5 | 1.2 | 0        |
| S1       | 0.6  | 1.0 | 2.1 | <b>1</b> |

$$d(S1, T1) = (0.5 - 0.6)^2 + (1.2 - 1.1)^2 + (2.3 - 2.1)^2 = \mathbf{0.06}$$

$$d(S1, T2) = 2.13$$

$$d(S1, T3) = 13.38$$

$$d(S1, T4) = 1.55$$

**K=1**  
**PREDICT**  
**CLASS 1**

# K-Nearest Neighbors

Example with Euclidean distance

| Examples | X1   | X2  | X3  | Label    |
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| T4       | -0.1 | 0.5 | 1.2 | 0        |
| S1       | 0.6  | 1.0 | 2.1 | <b>0</b> |

$$d(S1, T1) = (0.5 - 0.6)^2 + (1.2 - 1.1)^2 + (2.3 - 2.1)^2 = \mathbf{0.06}$$

$$d(S1, T2) = \mathbf{2.13}$$

$$d(S1, T3) = 13.38$$

$$d(S1, T4) = \mathbf{1.55}$$

**K=3**  
**PREDICT**  
**CLASS 0**

# Main Features



It is NOT a learning algorithm!

Despite being very simple, it may work well in practice when dealing with a large training set with a meaningful distance measure

...But beware of the following drawbacks:

- when dealing with a large training set :-)
- when dealing with high dimensional spaces

# Regression



- KNN can be used for regression
- Simply compute target value as (weighted) **combination** of those of the K nearest neighbors

$$\hat{y}_j = \frac{\sum_{k \in \mathcal{N}_j} w_k y_k}{\sum_{k \in \mathcal{N}_j} w_k}$$

$$w_k = \frac{1}{d(x_j, x_k)}$$

# Final Remarks



- It needs numeric features to compute distances
- **Feature scaling is a crucial element**
- It can be used for both classification and regression
- Neighbors can be weighted by their distance, so that closer neighbors count more
- Decide what to do in case of ties
- Curse of dimensionality in high-dimensional space